

Activity - 3

Analytical

Hough Circle Transform

1) Give two edge point

$$P_1 = (5, 4)$$

$$P_2 = (7, 6)$$

Centers $C_1 = (2, 4)$

$$C_2 = (4, 3)$$

P_1 to C_1

$$(5, 4) \quad (2, 4)$$

$$r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\sqrt{(5-2)^2 + (4-4)^2} = \sqrt{9+0}$$
$$= \sqrt{9} = 3$$

$$\boxed{r=3}$$

P_2 to C_1

$$\sqrt{(7-2)^2 + (6-4)^2} = \sqrt{25+4}$$
$$= \sqrt{29}$$

$$\boxed{r=\sqrt{29}}$$

P_1 to C_2

$$(5, 4) \text{ to } (4, 3)$$

$$\sqrt{(5-4)^2 + (4-3)^2} = \sqrt{2}$$

$$\boxed{r=\sqrt{2}}$$

P_2 to C_2

$$(7, 6) \quad (4, 3)$$

$$\sqrt{(7-4)^2 + (6-3)^2} = \sqrt{9+9}$$
$$r = \sqrt{18}$$

$$\boxed{r=\sqrt{18}}$$

No centre gives the maximum number of matching radius values.

2) Given $P(8,6)$
 $C(4,3)$

$$\text{Radius} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(4-8)^2 + (3-6)^2}$$

$$= \sqrt{16+9}$$

$$= \sqrt{25}$$

$$\boxed{r=5}$$

3) $C(5,4)$ $P(8,8)$

$$\text{Radius} = \sqrt{(8-5)^2 + (8-4)^2}$$

$$= \sqrt{3^2 + 4^2}$$

$$= \sqrt{9+16}$$

$$\boxed{r=5}$$